ICML 2024 | You Don't Have to Hand-Build Molecular Networks Anymore

Peking University and Microsoft Research AI4Science introduce GeoMFormer: two standard Transformer streams, bridged by cross-attention, that learn invariant and equivariant molecular representations at once.

Any deep-learning model for molecules has to obey a strict piece of physics: its predictions must transform the same way the coordinate system does. Rotate or translate the whole molecule in space, and its energy (a scalar) must not change at all, a property called invariance. But the force on each atom, or an atom's position (both 3D vectors), must rotate right along with it, a property called equivariance. The catch is that a single model is often asked to do both: predict the energy and the forces, the property and the structure. That means it has to learn invariant and equivariant representations well at the same time.

For years, the field has met these constraints by hand-building specialized geometric networks: some rely on expensive higher-order tensor operations, others stack tightly constrained equivariant modules. They are physically correct, but the price is architectures that are hard to scale, limited in expressive power, and usually good at only one side of the problem. It raises an obvious question. Instead of designing a new bespoke module every time, could we take the most standard, best-understood Transformer components and arrange them into a single framework that is general, flexible, and learns both kinds of representation well?

The ICML 2024 paper's answer is GeoMFormer, and its core idea is almost disarmingly simple. Keep two representations for every atom, one invariant and one equivariant, run them through two parallel standard Transformer streams, and connect the streams with carefully designed cross-attention modules so that invariant and equivariant information can consult each other's context. The authors go a step further and show that many previously proposed geometric networks are, in fact, special cases of this framework.

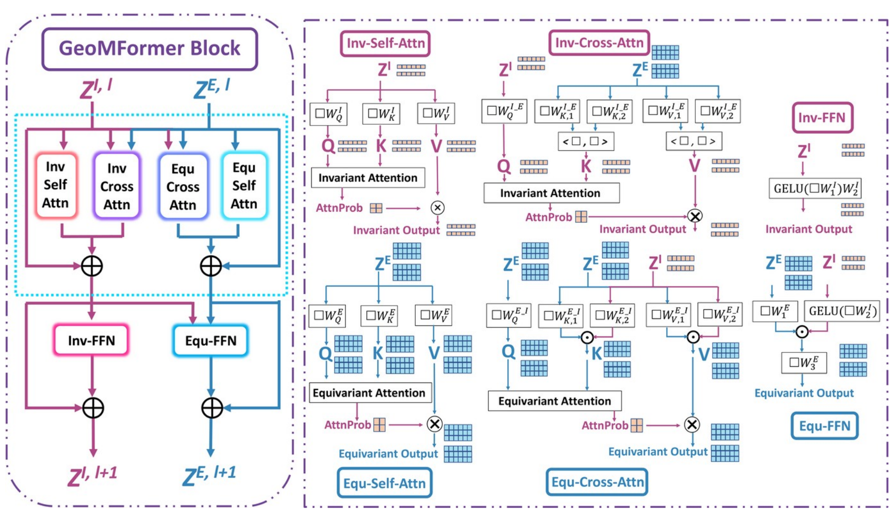

Figure 1. The GeoMFormer architecture. Left: one GeoMFormer Block, with the invariant stream (purple) and equivariant stream (blue) running in parallel, each passing through self-attention, a cross-attention module

reaching into the other stream, and a feed-forward network. Right: the four attention modules unfolded. The equivariant attention score is summed over the 3D Query and Key, which preserves equivariance by construction.

Paper: https://arxiv.org/abs/2406.16853 Code: https://github.com/c-tl/GeoMFormer

Two physical constraints, and why they are hard to satisfy together

Start with the problem itself. Scalar quantities such as energy or the HOMO-LUMO gap demand invariance: however you orient the molecule, the number you read off is the same. Vector quantities such as per-atom forces or relaxed atomic coordinates demand equivariance: rotate the molecule by 30 degrees and the predicted forces have to rotate by 30 degrees too. An ordinary neural network satisfies neither by default, and training one naively will not give you physically valid outputs.

The recurring weakness of earlier solutions is that they weld the constraint into a special-purpose module. Equivariant operations are inherently limited in number, because the equivariance constraint rules many operators out, so architectures grow complicated just to guarantee that one physical property. The result is models that are specialized, hard to reuse, compute-hungry on large systems, and typically strong at either invariant or equivariant prediction but not both. GeoMFormer takes a different route: rather than inventing new operators, it reorganizes the standard, thoroughly validated components of the Transformer.

The method: two standard Transformer streams, bridged by cross-attention

GeoMFormer keeps two tokens per atom: an invariant feature vector Z^I and an equivariant 3D feature Z^E. Each enters its own parallel Transformer stream. Inside a stream, self-attention first exchanges information across atoms. The key move is in the equivariant stream: standard attention is rewritten so the attention score is computed by summing dot products over the 3D Query and Key, channel by channel, which the authors prove keeps the output equivariant. The invariant stream uses ordinary scalar attention.

What actually ties the two streams together is the cross-attention module. The invariant stream queries the equivariant stream (Inv-Cross-Attn), and the equivariant stream queries the invariant stream in return (Equ-Cross-Attn); the two construct their Query/Key/Value differently, but each is designed so the fused output still respects its own geometric constraint. Intuitively, this lets scalar and vector information complement each other: an energy prediction can draw on geometric-structure cues, while force and position predictions can borrow the chemical semantics carried in the invariant features. Each block closes with a feed-forward network (Inv-FFN / Equ-FFN), combined with standard Layer Normalization and structural encodings, and the blocks are stacked.

Because the whole design is just a rearrangement of standard Transformer components, the authors can prove a striking consequence: by restricting or removing certain modules, a range of existing geometric networks, including Graphormer-3D, Equiformer, and TorchMD-Net, fall out as special cases of GeoMFormer. In other words, this is not simply one more model but a unifying framework that absorbs many prior methods.

The evidence: one architecture across invariant and equivariant tasks

The authors evaluate GeoMFormer on a suite deliberately built to test both sides at once. On OC20 (Open Catalyst 2020, over 460,000 adsorbate-catalyst complexes) they run both IS2RE (predict relaxed energy from the initial structure, invariant) and IS2RS (predict relaxed structure, equivariant). On PCQM4Mv2 (~3.37M molecules) and Molecule3D (~2.34M molecules) they predict the HOMO-LUMO gap. A five-particle N-body simulation adds an equivariant position-prediction test. Across all of them, the same architecture reaches or approaches the state of the art.

Table 1. GeoMFormer versus the previous best baseline on each task, spanning both invariant and equivariant prediction. Arrows give the metric direction: ↓ lower is better, ↑ higher is better.

Task / Metric	GeoMFormer	Previous best baseline
OC20 IS2RE, Energy MAE (eV) ↓	0.4141	0.4410 (Equiformer)
OC20 IS2RS, ADwT (%) ↑	10.67	8.36 (GemNet-OC)
PCQM4Mv2, Valid MAE ↓	0.0734	best among $O(n^2)$ models, −6.7% rel.
Molecule3D random split, MAE ↓	0.0252	0.0301 (SphereNet)
Molecule3D scaffold split, MAE ↓	0.1045	0.1182 (SphereNet)
N-body simulation, MSE ↓	0.0047	0.0071 (EGNN)

A few numbers are worth singling out. On PCQM4Mv2, GeoMFormer reaches a validation MAE of 0.0734, the lowest of any $O(n^2)$ (quadratic-complexity) model, a 6.7% relative reduction over the previous best quadratic model while staying efficient enough to scale. On Molecule3D it improves MAE by 16.3% on the random split and 11.6% on the scaffold split. And on the N-body simulation it cuts MSE from 0.0071 to 0.0047, a 33.8% relative reduction. One model beats specialized invariant models on energy-style tasks and leads on structure- and position-style equivariant tasks.

Ablation: cross-attention is the heart of the design

It is natural to ask where the gains come from. The authors ablate each of the four attention modules, and the most telling result centers on the cross-attention that connects the two streams: remove it and performance drops sharply. On the N-body simulation, dropping either cross-attention module on its own already costs a meaningful amount, and dropping both together hurts most.

Table 2. Ablating the cross-attention modules on the N-body simulation (all other hyperparameters held fixed). Values are the performance drop relative to the full model; a larger negative number means the model depends more on that module.

Configuration	Performance drop
Full model (all four attention modules)	baseline
Remove Inv-Cross-Attn	−8.5%
Remove Equ-Cross-Attn	−14.9%
Remove both cross-attention modules	−21.3%

On the harder MD17 force-field task the effect is even larger: take out that bridge and the error climbs several-fold. Self-attention, the feed-forward network, and Layer Normalization each contribute too, but the module that genuinely fuses the invariant and equivariant streams is cross-attention. That squares with the paper's central claim: letting the two geometric representations see each other is what makes the framework work.

Takeaway: standard components can cover geometric modeling

The message of GeoMFormer is clear. Modeling molecules under physical constraints does not necessarily require a pile of one-off, heuristic modules. Run two parallel standard Transformer streams, one for invariant and one for equivariant features, connect them with simple cross-attention, and a single model learns both kinds of representation at once while outperforming specialized architectures across a broad set of tasks. Because many existing methods emerge as special cases, GeoMFormer reads less like yet another model and more like a clean, general, scalable design principle for geometric molecular representation learning.

The boundaries are worth stating plainly. The attention here is $O(n^2)$, and the paper's claim is best among quadratic-complexity models; a linear-time version for very large systems remains future work. The empirical study covers energy, forces, structure, and particle positions, but it is aimed at molecular and particle systems, so transfer to other geometric domains still has to be demonstrated. Even so, the simple two-streams-plus-cross-attention pattern gives follow-up work a starting point well worth building on.